Transferable Adversarial Perturbation Generator: A Lightweight Robustness Evaluation Tool for ML Engineers

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Abstract

Modern vision models achieve remarkable accuracy on clean benchmarks yet remain vulnerable to small, often imperceptible perturbations that cause misclassification. While iterative white-box and query-based black-box attacks are well-studied and broadly implemented in libraries like ART and Foolbox, they are not designed as lightweight, deployment-ready tools that emphasize transferability across models and domains. We present a practical system for ML engineers: a transferable adversarial perturbation generator paired with a usability-focused robustness dashboard deployed on Streamlit (https://adv-toolkit.streamlit.app). Our approach implements two complementary, low-compute strategies: (1) the ultra-light Sine Attack (SA) that parameterizes perturbations via sinusoidal bases, and (2) a cross-domain perturbation generator optimized through Random Normalization and Domain Agnostic Attention module. We target a streamlined pipeline where users upload a model and images, upload optional custom transform json, click Run Attack, and receive comprehensive metrics including attack success rate, fooling rates, perturbation norms (L_0, L_2, L_∞), and perceptual similarity measures (SSIM, PSNR, LPIPS). Unlike existing toolboxes that require extensive querying or white-box access, our transfer-based approach operates in realistic black-box scenarios without multiple queries. By packaging recent insights on transferable and cross-domain attacks into an accessible tool with downloadable metrics and adversarial examples, we lower the barrier to proactive robustness checks during model development and deployment.

1. Introduction

Deep neural networks (DNNs) can be brittle: small, structured perturbations—often imperceptible to humans—can flip predictions [6, 17]. These failure modes rarely appear on clean validation sets and are easy to overlook without targeted testing. In practical deployment, a model that performs well on held-out data may still be compromised by simple transfer attacks that exploit cross-model vulnerabilities. Current adversarial toolboxes (*e.g.*, IBM ART [11], Foolbox [16]) provide comprehensive attack implementations but have significant limitations for practical deployment scenarios:

- Zero-query black-box evaluation that doesn't require target model access
- Comprehensive metrics beyond simple accuracy: fooling rates, multiple perturbation norms, perceptual similarity scores
- Flexible JSON-based transform specification instead of hard-coded preprocessing
- Downloadable results including metrics JSON and adversarial example archives
- Web-based dashboard requiring no local installation (Streamlit deployment)

A user can upload a trained model or dataset, run a fast evaluation that produces transferable perturbations, and receive quantitative indicators of vulnerability through an intuitive interface. By emphasizing transferability, this tool mimics realistic black-box threat scenarios and reveals weaknesses that traditional accuracy metrics may obscure.

1.1. Contributions

This work makes the following contributions:

1. A low-compute implementation and evaluation of two transfer-oriented attacks: Sine Attack (SA) that parameterizes perturbations via frequency-domain sinusoidal bases, and Beyond ImageNet Attack (BIA) that generates trasnferable pertubations optimized via Random Nornmalization and Domain Agnostic Attention module. [20]
2. An ensemble-based attack fusion strategy that combines frequency-domain (SA) and pixel-space (BIA) perturbations, providing robustness when individual attacks fail.
3. A comprehensive evaluation framework with eight metrics: attack success rate, model clean accuracy, fooling

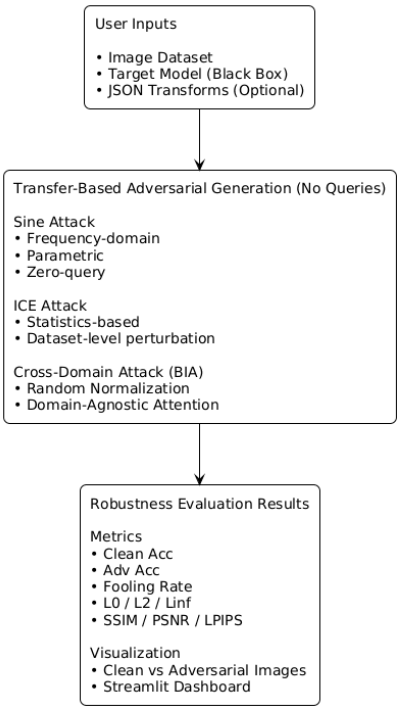


Figure 1. System overview of our transfer-based adversarial robustness evaluation toolkit. Users provide a trained target model, evaluation images, and optional preprocessing transforms. Adversarial examples are generated using zero-query, transfer-based attacks, including frequency-domain, statistics-based, and cross-domain methods. The system reports comprehensive robustness metrics and visualizations through an interactive Streamlit dashboard.

rate, three perturbation norms (L_0, L_2, L_∞), and three perceptual similarity measures (SSIM, PSNR, LPIPS).

4. A novel transfer-based adversarial robustness toolkit with a plug-and-play dashboard (`https://adv-toolkit.streamlit.app`) that provides JSON-configurable transforms, downloadable metrics, and adversarial example archives—all without requiring local installation or target model queries.

2. Background

2.1. Adversarial Examples and Threat Models

Early work by Szegedy *et al*. [17] and Goodfellow *et al*. [6] revealed the existence of small-ℓ_∞ or ℓ_2-bounded perturbations that cause misclassification in neural networks. These adversarial examples exploit the high-dimensional nature of input spaces and the locally linear behavior of neural networks. Formally, given a classifier $f : \mathcal{X} \to \mathcal{Y}$, input x with true label y, and perturbation budget ϵ, an adversarial example x' satisfies: $\|x' - x\|_p \leq \epsilon$ and $f(x') \neq y$, where $\|\cdot\|_p$ denotes the L_p norm.

Adversarial attack methods are typically categorized by their threat model—the information available to the attacker. **White-box attacks** assume complete access to model architecture, weights, and gradients, enabling optimization-based methods like PGD [8] and C&W [2]. **Black-box attacks** operate with limited information and are further divided into two categories:

- **Query-based attacks:** The attacker can query the target model (potentially thousands of times) to observe outputs or confidence scores. Methods like HopSkipJump [3], Square Attack [1], and ZOO [4] estimate gradients or decision boundaries through strategic probing. While effective, these approaches require extensive interaction with the target system (often 1,000–10,000 queries per image), making them:
  - Easily detectable through rate limiting or anomaly detection
  - Impractical for rapid evaluation of large test sets
  - Expensive when targeting commercial APIs with per-query pricing
  - Slow, requiring minutes to hours per image
- **Transfer-based attacks:** The attacker generates adversarial examples using surrogate (substitute) models without any queries to the target. These attacks exploit the transferability phenomenon—adversarial examples crafted for one model often fool other models, even with different architectures or training procedures [12, 18]. Transfer-based attacks offer significant advantages:
  - Zero queries required, making attacks undetectable
  - Fast evaluation, processing images in seconds
  - Realistic threat model for deployed systems
  - Suitable for proactive robustness testing during development

2.2. Transferability and Universal Perturbations

Adversarial examples exhibit remarkable transferability: perturbations optimized against one model frequently cause misclassification in other models, even with different architectures, training procedures, or data augmentation strategies [12]. This phenomenon suggests that adversarial examples exploit common vulnerable directions in the input space shared across models. Papernot *et al*. [13] demonstrated that transferability enables practical black-box attacks through surrogate model training.

Universal adversarial perturbations (UAPs) [9] extend this concept by learning image-agnostic perturbation vectors δ that fool diverse inputs from a distribution. Formally, a UAP seeks a perturbation δ satisfying $\|\delta\|_p \leq \epsilon$ such that for most samples $x \sim \mathcal{D}$: $f(x + \delta) \neq f(x)$. UAPs reveal that models share systematic vulnerabilities across their input space, not just for specific examples.

Cross-domain transferability [10] investigates whether perturbations trained on one data distribution can attack

models trained on different domains. This is particularly relevant for deployment scenarios where attackers may not have access to data from the target distribution. Naseer *et al*. [10] showed that perturbations can transfer across visually distinct domains (*e.g*., natural images, sketches, paintings), suggesting domain-invariant vulnerable features in neural networks.

2.3. Frequency-Domain Analysis of Adversarial Perturbations

Recent work has analyzed adversarial perturbations through the lens of frequency decomposition, revealing that CNNs are particularly sensitive to specific frequency bands. Yin *et al*. [19] demonstrated that adversarial perturbations typically contain high-frequency components that exploit CNN sensitivities beyond human perception. Geirhos *et al*. [5] showed that ImageNet-trained CNNs are biased toward texture (high-frequency) rather than shape (low-frequency) information, making them vulnerable to texture-based attacks.

Qin *et al*. [15] discovered that simple sinusoidal perturbations with specific frequencies (0.15–0.24 Hz) and orientations (45° and 135°) can effectively fool CNNs across datasets. Their Sine Attack (SA) parameterizes perturbations as:

$$Z_j(h, w) = \sin(a_j \cdot h + b_j \cdot w + c_j), \quad j \in \{R, G, B\} \quad (1)$$

where a_j, b_j control frequency and direction, and c_j controls phase. This compact representation (only 9 parameters for RGB images) enables efficient optimization while maintaining strong transferability. The success of SA suggests that CNNs have intrinsic vulnerabilities to periodic patterns in specific frequency bands.

2.4. Generative Adversarial Perturbations

Instead of per-image optimization, generative approaches train neural networks to produce perturbations quickly at inference time. Generative Adversarial Perturbations (GAP) [14] and subsequent methods reduce run-time cost while maintaining attack strength. These approaches typically train a generator network G that takes an input image x and produces a perturbation $\delta = G(x)$, optimized to maximize classification error on surrogate models.

Li *et al*. [7] propose **Beyond ImageNet Attack (BIA)** to address cross-domain transferability, where adversarial perturbations are trained on a source domain (e.g., ImageNet) but must generalize to models trained on different visual distributions. Unlike conventional attacks that assume shared data and preprocessing distributions, BIA explicitly targets unknown target domains through two key mechanisms.

Random Normalization (RN). A major obstacle to cross-domain transfer is inconsistent input normalization across models. Perturbations optimized under ImageNet statistics often fail when deployed against models using different preprocessing pipelines. RN mitigates this by randomly sampling normalization parameters during training:

$$\mathcal{RN}(x_s) = \sigma \cdot \frac{x_s - \mu'}{\sigma'} + \mu, \quad (2)$$

where x_s is the source image, (μ', σ') denote source-domain normalization, and (μ, σ) are randomly sampled from common target normalization schemes. This forces the generator to learn perturbations that are *normalization-agnostic*, substantially improving transferability to models with unknown preprocessing.

Domain-Agnostic Attention Module (DAAM). Standard attention mechanisms often overfit to domain-specific visual cues, limiting cross-domain generalization. DAAM instead emphasizes features that are stable across domains by applying cross-channel average pooling to intermediate feature maps (e.g., $Maxpool_3$ in VGG-16 and $Conv3_8$ in DCL with a ResNet-50 backbone). The generator $\mathcal{G}_\theta$ is trained using a domain-agnostic attention loss:

$$\theta^* = \arg\min_\theta \mathcal{L}_{\cos}\left(\mathcal{A}^L \odot f_s^L(x_s), \mathcal{A}^L \odot f_s^L(x_s)\right), \quad (3)$$

where $\mathcal{A}^L$ denotes attention weights at layer L, f_s^L are source-model features, and $\odot$ is the Hadamard product. By prioritizing channel-wise, domain-invariant statistics rather than spatial patterns, DAAM guides the generator toward perturbations that exploit vulnerabilities shared across diverse visual domains.

Together, RN and DAAM enable BIA to achieve strong cross-domain transferability under significant uncertainty in target model architecture, training data, and preprocessing, making it well suited for realistic black-box attack settings.

2.5. Existing Adversarial Robustness Toolboxes

Several open-source libraries provide adversarial attack implementations. IBM's Adversarial Robustness Toolbox (ART) [11] and Foolbox [16] offer comprehensive attack collections, primarily focusing on iterative optimization methods and query-based approaches. However, these tools have limitations for practical deployment:

- Focus on research-grade implementations requiring expert knowledge
- Lack of user-friendly interfaces for model developers and ML engineers
- Limited emphasis on transfer-based attacks for zero-query scenarios
- No integrated evaluation dashboard with comprehensive metrics
- Require transforms to be embedded in model code rather than flexible configuration
- No support for cross-domain robustness evaluation

Our toolkit addresses these gaps by providing a deployment-ready system focused on transfer-based attacks with an intuitive web interface, comprehensive metrics, and flexible configuration options.

2.6. Implementation Details

We implement two complementary transfer-based attack strategies, designed to be fast and generalize across models and domains. Both methods operate without querying the target model, making them suitable for realistic black-box scenarios.

2.7. Sine Attack: Ultra-Light Parametric Perturbations

Following Qin *et al*. [15], we implement the Sine Attack (SA) which parameterizes perturbations as sums of channel-wise sine waves with learnable parameters. The key insight is that CNNs are systematically vulnerable to periodic patterns in specific frequency bands (0.15–0.24 Hz) and orientations (45° and 135°). SA exploits this vulnerability through a compact representation requiring only 9 parameters for RGB images.

2.7.1. Formulation

Sine Attack (SA) constructs adversarial perturbations using a compact, frequency-based parameterization inspired by the spectral structure observed in generalized transferable attacks. Rather than optimizing per-pixel noise or a neural generator, SA represents the perturbation for each color channel as a single 2D sinusoidal signal.

Formally, for an input image of spatial resolution $H \times W$, the perturbation applied to channel $j \in \{R, G, B\}$ is defined as

$$Z_j(h, w) = \sin(a_j \cdot h + b_j \cdot w + c_j), \tag{4}$$

where (h, w) denote the pixel coordinates with $h \in \{1, \ldots, H\}$ and $w \in \{1, \ldots, W\}$. The scalars a_j and b_j control the spatial frequency and orientation of the sine wave, while c_j determines its phase offset. Each channel is therefore parameterized by three learnable scalars, resulting in a total of nine parameters for RGB images.

The full perturbation tensor is constructed by concatenating the channel-wise sine maps:

$$Z(\omega) = \text{Concat}\left(Z_R, Z_G, Z_B\right), \tag{5}$$

where $\omega = \{a_R, b_R, c_R, a_G, b_G, c_G, a_B, b_B, c_B\}$ denotes the complete parameter vector.

Given a clean input image x, the adversarial example is generated as

$$\hat{x} = \text{Clip}\left(x + \epsilon \cdot \text{sign}\left(Z(\omega)\right)\right), \tag{6}$$

where ϵ specifies the L_∞ perturbation budget and $\text{Clip}(\cdot)$ enforces valid pixel ranges.

This formulation yields several desirable properties:

- The perturbation is resolution-agnostic, as the sine parameters scale naturally with image dimensions.
- The attack has an extremely low parameter count, enabling fast optimization and inference.
- The structured, periodic nature of the perturbation targets intrinsic frequency-domain vulnerabilities shared across CNN architectures.

Empirically, effective sine parameters concentrate within specific frequency bands (approximately 0.15–0.24 Hz) and diagonal orientations (45° and 135°), consistent with prior spectral analyses of transferable adversarial noise [15]

2.7.2. Training Procedure

We train SA on Tiny-ImageNet using the first 20 classes with ResNet-18 and MobileNetV2 as surrogate models. The objective is to maximize the cross-entropy loss on the surrogate ensemble:

$$\max_{\omega} \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \mathbb{E}_{(x,y)\sim\mathcal{D}}[\text{CE}(f_m(x + \epsilon \cdot \text{sign}(Z(\omega))), y)] \tag{7}$$

where $\mathcal{M}$ is the surrogate model ensemble, $\mathcal{D}$ is the training distribution, and CE denotes cross-entropy loss. The gradient with respect to ω is computed as:

$$\nabla_\omega l = \epsilon \cdot \nabla_{\hat{x}} \text{CE}(f(\hat{x}), y) \cdot \nabla_\omega \text{Clip}(x + \epsilon \cdot Z(\omega)) \tag{8}$$

We optimize ω using AdamW with learning rate 0.01 for 15 epochs with batch size 32. Once trained, SA generates perturbations instantaneously at inference time—requiring only sine wave evaluation and no gradient computation.

2.7.3. Advantages

- Extreme computational efficiency: 5 parameters vs. millions for generative networks
- Instant inference: no forward pass through neural networks required
- Strong cross-model transferability due to exploitation of fundamental CNN vulnerabilities
- Resolution-agnostic: same parameters work across different image sizes

2.8. Beyond ImageNet Attack: Cross-Domain UAP

Our second approach uses the Beyond ImageNet Attack [7], a learned generator network specifically designed for cross-domain transferability. Unlike traditional attacks that assume source and target models are trained on the same distribution, Beyond ImageNet Attack addresses the realistic scenario where attackers train on ImageNet but target models trained on different domains.

2.8.1. Architecture

The generator $G : \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^{H\times W\times 3}$ takes an input image x and produces a perturbation $\delta = G(x)$. The architecture incorporates two key innovations:

Random Normalization (RN): During training, input images are randomly normalized to simulate different preprocessing pipelines:

$$\mathcal{RN}(x_s) = \sigma \cdot \frac{x_s - \mu'}{\sigma'} + \mu \tag{9}$$

where x_s is the source image, μ' and σ' are source domain normalization parameters, and μ, σ are randomly sampled from distributions of common normalization schemes. This ensures the generator produces perturbations robust to varying preprocessing across target models.

Domain-Agnostic Attention Module (DAAM): A specialized attention mechanism that learns to focus on domain-invariant features. Cross-channel average pooling is applied to intermediate feature maps from surrogate models (e.g., $Maxpool_3$ of VGG-16). The optimization objective with domain-agnostic attention is:

$$\theta^* = \arg\min_{\theta} \mathcal{L}_{cos}(\mathcal{A}^L \odot f_s^L(x_s), \mathcal{A}^L \odot f_s^L(x_s)) \tag{10}$$

where $\odot$ is Hadamard product, $\mathcal{A}^L$ are attention weights, and f_s^L are feature maps at layer L. This guides the generator to produce perturbations that exploit universal vulnerabilities rather than domain-specific artifacts.

2.8.2. Training Strategy

We use the pre-trained ResNet-18 based generator from the Beyond ImageNet Attack repository, trained against VGG-19 surrogate model on ImageNet and heterogeneous datasets with the following objective:

$$\max_G \mathbb{E}_{(x,y)\sim\mathcal{D}_{\text{ImageNet}}}[\mathcal{L}_{\text{adv}}(f_{\text{VGG19}}(\mathcal{RN}(x)+G(\mathcal{RN}(x))), y)] \tag{11}$$

where the adversarial loss $\mathcal{L}_{\text{adv}}$ combines cross-entropy with regularization:

$$\mathcal{L}_{\text{adv}} = \text{CE}(f(\hat{x}), y)+\lambda_2 \cdot \|G(x)\|_2+\lambda_{TV} \cdot \text{TV}(G(x)) \tag{12}$$

where TV denotes total variation (promoting spatial smoothness). The pre-trained weights were optimized with $\lambda_2 = 0.1$ and $\lambda_{TV} = 0.01$.

2.8.3. Inference

At test time, given an input image x, we generate the adversarial example as:

$$\hat{x} = \text{Clip}(x + \epsilon \cdot \text{sign}(G(x))) \tag{13}$$

The sign projection ensures the perturbation is ϵ-bounded in L_∞ norm while the generator provides pixel-space perturbations that exploit learned vulnerabilities. A single forward pass through the ResNet-18 generator takes approximately 10–20ms on GPU, making it suitable for rapid evaluation.

2.9. Ensemble Attack Strategy

To maximize robustness evaluation coverage, we implement an ensemble strategy that combines both attack methods. SA and the Beyond ImageNet generator exploit different vulnerability types:

- SA targets frequency-domain vulnerabilities with periodic patterns
- Beyond ImageNet exploits learned pixel-space features with domain-agnostic attention

Our ensemble approach applies both attacks and prefers BIA if both fooled or falls back on Sine Attack. This provides complementary coverage: when one attack fails to fool a model (*e.g.*, due to frequency-domain defenses), the other may succeed. The evaluation pipeline computes metrics for the combined attacks.

2.10. Evaluation Metrics

We report a comprehensive suite of eight metrics to characterize attack effectiveness and perturbation quality:

2.10.1. Attack Effectiveness Metrics

- **Attack Success Rate (ASR)**: Percentage of correctly classified clean images that are misclassified after perturbation. Formally:

$$\text{ASR} = \frac{|\{x : f(x) = y \wedge f(\hat{x}) \neq y\}|}{|\{x : f(x) = y\}|} \tag{14}$$

  where f is the target classifier, $\hat{x}$ is the adversarial example, and y is the true label.
- **Model Clean Accuracy**: Classification accuracy on unperturbed test images, providing baseline performance. This contextualizes ASR—a model with 95% clean accuracy where ASR=50% indicates the attack successfully fooled more than half of correctly classified images.
- **Fooling Rate (FR)**: Percentage of all test images (including initially misclassified) that change predicted label after perturbation:

$$\text{FR} = \frac{|\{x : f(\hat{x}) \neq f(x)\}|}{|\mathcal{D}_{\text{test}}|} \tag{15}$$

2.10.2. Perturbation Magnitude Metrics

- L_0 **norm**: Number of modified pixels. $L_0 = |\{(i,j) : \hat{x}_{ij} \neq x_{ij}\}|$. Lower values indicate more localized perturbations.
- L_2 **norm**: Euclidean distance between clean and adversarial images. $L_2 = \|\hat{x} - x\|_2 = \sqrt{\sum(\hat{x}_{ij} - x_{ij})^2}$. Commonly used perturbation budget metric.
- L_∞ **norm**: Maximum per-pixel perturbation. $L_\infty = \|\hat{x} - x\|_\infty = \max_{ij} |\hat{x}_{ij} - x_{ij}|$. This is our primary constraint during attack generation ($\epsilon = 8/255$ for CIFAR, $\epsilon = 16/255$ for ImageNet).

2.10.3. Perceptual Similarity Metrics

- **Structural Similarity Index (SSIM)**: Measures perceived structural similarity between images, ranging from -1 to 1 (1 = identical). SSIM considers luminance, contrast, and structure:

$$\text{SSIM}(x, \hat{x}) = \frac{(2\mu_x\mu_{\hat{x}} + C_1)(2\sigma_{x\hat{x}} + C_2)}{(\mu_x^2 + \mu_{\hat{x}}^2 + C_1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + C_2)} \quad (16)$$

  where μ, σ denote means and standard deviations. Higher SSIM indicates better perceptual similarity.
- **Peak Signal-to-Noise Ratio (PSNR)**: Measures ratio between maximum signal power and perturbation noise:

$$\text{PSNR} = 20 \cdot \log_{10}(\text{MAX}) - 10 \cdot \log_{10}(\text{MSE}) \quad (17)$$

  where MAX is the maximum pixel value and MSE is mean squared error. Higher PSNR indicates less perceptible perturbation (typically > 30 dB is considered good quality).
- **Learned Perceptual Image Patch Similarity (LPIPS)**: Deep learning-based perceptual metric using VGG features:

$$\text{LPIPS}(x, \hat{x}) = \sum_l w_l \cdot \|\phi_l(x) - \phi_l(\hat{x})\|^2 \quad (18)$$

  where ϕ_l denotes features from layer l of VGG network and w_l are learned weights. Lower LPIPS indicates better perceptual similarity. LPIPS correlates better with human perception than PSNR or SSIM [21].

These eight metrics provide comprehensive characterization: attack effectiveness (ASR, clean accuracy, fooling rate), perturbation magnitude (L_0, L_2, L_∞), and perceptual quality (SSIM, PSNR, LPIPS). Our dashboard reports all metrics and exports them in JSON format for further analysis.

2.11. JSON-Configurable Transforms

A key innovation of our toolkit is flexible, JSON-based transform specification. Unlike existing toolboxes (ART, Foolbox) that require transforms to be embedded within model code or specified programmatically, our system accepts transform configurations as JSON files.

Example JSON transform specification:

```
{
  "transforms": [
    {
      "type": "Resize",
      "params": {"size": [224, 224]}
    },
    {
      "type": "Normalize",
      "params": {
        "mean": [0.485, 0.456, 0.406],
        "std": [0.229, 0.224, 0.225]
      }
    }
  ]
}
```

The system automatically parses the JSON and constructs a PyTorch transform pipeline, applying it consistently to both clean and adversarial images during evaluation. Supported transforms include: Resize, CenterCrop, Pad, Normalize, Grayscale, ConvertImageDtype.ToTensor is applied before passing the tensors through the custom transform.

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3. Datasets

We evaluate our system using standard vision benchmarks commonly employed in adversarial robustness research.

CIFAR-10. CIFAR-10 contains 50,000 training images and 10,000 test images across 10 classes, with resolution 32×32. CIFAR-10 is used for training surrogate models, computing ICE statistics, and validating end-to-end attack integration. All 10 classes are observed during training, as confirmed by ICE statistics.

Tiny-ImageNet. Tiny-ImageNet consists of 200 classes with 64×64 RGB images. A 20-class subset is used for training and evaluating the Sine Attack parameters, following prior work that demonstrates strong transferability even under partial class coverage.

ImageNet (Pretrained Weights). For cross-domain attacks, we rely on pretrained Beyond ImageNet Attack (BIA) generator weights trained on ImageNet-scale data. No ImageNet training is performed within our pipeline; evaluation focuses on transfer behavior rather than retraining.

All datasets are used strictly under a transfer-based black-box threat model, with no queries made to target models during attack generation.

4. Initial Experimental Results

We report SOME initial experimental results obtained from experiments prior to integration into the Streamlit application. These experiments validate correctness, feasibility, and system behavior rather than state-of-the-art performance.

4.1. ICE Attack Statistics

We evaluate ICE-style perturbations on CIFAR-10 using two target models: ResNet-18 (Model 0) and MobileNetV2 (Model 1). Training statistics confirm full dataset coverage:

- Samples processed: 50,000
- Classes observed: 10/10

- Epochs: 1

Evaluation results are summarized below:

Model	Clean Acc.	Adv. Acc.	Fooling Rate (%)
ResNet-18	11.92%	11.46%	-1.86
MobileNetV2	9.40%	9.98%	6.17

Table 1. ICE attack evaluation on CIFAR-10. Fooling Rate measures prediction changes after perturbation.

These results indicate modest but measurable transfer-based impact without querying target models. Negative fooling rate for MobileNetV2 highlights the variability inherent to transfer-based attacks and motivates multi-attack evaluation.

4.2. Sine Attack Statistics

We evaluate the Sine Attack (SA) on CIFAR-10 using two target models: ResNet-18 and MobileNetV2. The attack is evaluated under a strict transfer-based black-box setting, with no access to target model parameters or gradients.

Final evaluation results on the test set are summarized below:

Model	Clean Acc.	Adv. Acc.	Fooling Rate (%)
ResNet-18	72.11%	71.43%	9.40
MobileNetV2	60.42%	58.43%	25.76

Table 2. Sine Attack evaluation on CIFAR-10 under a transfer-based black-box setting.

The results indicate that Sine Attack induces modest accuracy degradation while achieving non-trivial fooling rates, particularly on MobileNetV2. The higher fooling rate on MobileNetV2 suggests increased sensitivity to frequency-domain perturbations, despite a relatively small drop in overall classification accuracy.

These findings align with prior observations that frequency-based adversarial patterns can transfer across models while remaining visually subtle and computationally inexpensive to generate.

4.3. Cross-Domain Attack Statistics (Beyond ImageNet)

We additionally evaluate cross-domain adversarial perturbations using the Beyond ImageNet Attack (BIA) framework. These experiments are conducted to validate the generation and perceptual quality of perturbations under domain shift, rather than to optimize fooling rates.

The cross-domain attacker incorporates Random Normalization and a Domain-Agnostic Attention Module (DAAM), enabling perturbation generation that is robust to unknown target preprocessing and visual distributions. Experiments confirm successful end-to-end execution of the attack pipeline, including model loading, perturbation generation, and compatibility with TorchScript-exported target models.

Perturbation quality metrics indicate that cross-domain adversarial examples remain visually subtle. In particular, BIA perturbations achieve high perceptual similarity (SSIM ≈ 0.996), high PSNR (≈ 44 dB), and low LPIPS ($\approx 4 \times 10^{-4}$), comparable to those produced by AutoAttack under similar L_∞ budgets.

These results demonstrate that the cross-domain attack produces valid, low-distortion adversarial examples suitable for black-box robustness evaluation, even when the target model's training domain and preprocessing pipeline are unknown.

5. Experiments

Our experiments aim to validate system functionality, transfer-based attack behavior, and usability rather than benchmark optimal attack performance.

5.1. Target Models

We evaluate attacks against models not used during surrogate training:

- ResNet-18
- MobileNetV2

All models are treated as black boxes during evaluation, with no access to gradients, logits beyond predicted labels, or internal representations.

5.2. Evaluation Protocol

For each model, we compute:

- Clean accuracy
- Adversarial accuracy
- Fooling rate
- Perturbation norms (L_0, L_2, L_∞)
- Perceptual similarity metrics (SSIM, PSNR, LPIPS)

Metrics are computed uniformly across attacks and exported automatically through the Streamlit interface.

6. Training Details

All training is performed offline prior to deployment.

ICE Statistics Training. ICE statistics are computed using a single epoch over CIFAR-10 with batch size 128, covering all 50,000 samples and all 10 classes. This step serves to validate data coverage and attack readiness rather than to optimize performance.

Sine Attack. Sine Attack parameters are trained on Tiny-ImageNet using surrogate CNNs. Due to its parametric nature, SA requires negligible storage and no neural network inference at test time.

Beyond ImageNet Attack. We use publicly released pretrained BIA generator weights without modification. No additional fine-tuning is performed.

Runtime. SA perturbations are generated instantaneously, while BIA requires a single forward pass (10–20 ms on GPU), enabling rapid batch evaluation.

7. Streamlit Application

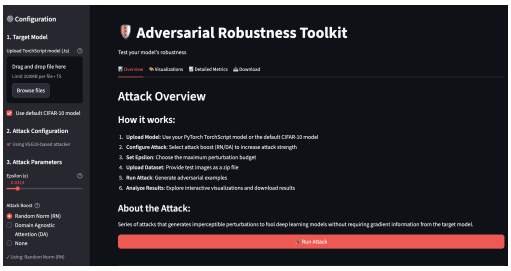
We deploy our system as a web-based Streamlit application (https://adv-toolkit.streamlit.app) to support interactive robustness evaluation.

7.1. User Workflow

Users interact with the system through the following steps:

1. Upload or select a trained model
2. Upload evaluation images or a dataset
3. (Optional) Upload JSON-based transform configuration
4. Run attacks and view results

A demo for the same is available **here**. It will also present comparisons and statistics as highlighted in the next section.



7.2. Outputs

The application provides:

- Interactive metric summaries
- Side-by-side clean and adversarial image comparisons
- Downloadable JSON metric reports
- Archived adversarial examples

The system requires no local installation and is designed for ML engineers without adversarial ML expertise.

8. Discussion

Our results demonstrate that transfer-based adversarial evaluation can be integrated into a lightweight, engineer-facing system. While individual attacks may exhibit limited fooling rates, their value lies in exposing model sensitivity under realistic black-box conditions.

We emphasize that our goal is not to achieve state-of-the-art attack performance, but to validate an end-to-end robustness evaluation workflow. Variability across models, including negative fooling rates, reflects the inherent uncertainty of transfer-based attacks rather than system failure.

9. Conclusions

We present a practical adversarial robustness evaluation toolkit combining frequency-domain and cross-domain transfer attacks with an accessible Streamlit interface. Our system demonstrates end-to-end functionality, meaningful perturbation generation, and comprehensive metric reporting under realistic threat models.

This work prioritizes usability and deployment realism over attack optimization and serves as a foundation for future extensions in transfer-based robustness testing.

Team & Author Contributions

Aakanksha Dutta implemented the Streamlit-based dashboard and conducted experiments involving the cross-domain adversarial attacker.

Zachary Tan developed the demonstration pipeline and performed experimental evaluation of the Sine Attack.

Aryan Raj Dhawan carried out paper writing, and conducted experiments using the ICE attack framework.

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